

Raster image type

1. **Binary:** is a digital image that has only two possible values for each pixel.
2. **Grayscale:** is an image in which the value of each pixel is a single sample, that is, it carries only intensity information.
3. **Color:** is a digital image that includes color information for each pixel.

Lightness: perceived achromatic intensity of reflecting object

Brightness: perceived intensity of a self-luminous object, such as a light bulb, the sun, or an LCD screen

Subtractive mixture

occurs when mixing paints, dyes, inks, etc. that act as a filters between the viewer and the light source / reflective surface.

2-Additive Mixture للشاشات

- *Additive mixture* occurs when color is created by mixing visible light emitted from light sources

Complementary hues: Any hue will approach **gray** if additively mixed with its opposite hue on the color

CMY AND **CMYK** COLOR MODELS

- **CMY:** Secondary colors of light, or primary colors of pigments are used.
- Used to generate hardcopy output (Printer and Copier).
- Some facts:
 - Printer papers are white (reflect all colors)
 - Printers use ink (Transparent)
- **K (Black)** is practical problem of $C+M+Y \neq \text{Black}$ (Muddy Brown) وذلك بسبب مشكلة في جودة الاحبار الكيميائية. Add a fraction of Black color
- **Hue**
- In HSV, hue represents color. In this model, hue is an angle from **0 degrees to 360 degrees**.

■ **Saturation**

- **Saturation** indicates the range of grey in the color space. It ranges from 0 to 100%. Sometimes the value is calculated from 0 to 1. When the value is '0,' the color is grey and when the value is '1,' the color is a primary color. A faded color is due to a lower saturation level, which means the color contains more grey.

■ **Value**

- **Value** is the brightness of the color and varies with color saturation. It ranges from 0 to 100%. When the value is '0' the color space will be totally black. With the increase in the value, the color space brightness up and shows various colors.

- **Warm colors** contain shades of yellow and red;
- **cool colors** have a blue, green, or purple tint;
- **Neutral colors** include brown, gray, black, and white.

1-Monochromatic

- ▶ all colors have hues that are the same or within a few degrees of one another

Complementary Colors are the colors opposite from one another on the wheel.

3-Analogous

- ▶ three colors which lie close together on a color wheel

4- Triadic

- ▶ Any 3 colors, spaced equally around a color wheel

5- Tetradic رباعي

- ▶ Four colors that are evenly spaced on the color wheel. Tetradic color schemes are bold and work best if you let one color be dominant الغالب, and use the others as accents مساعدين. The more colors you have in your palette, the more difficult it is to balance,

6- Split complementary

- ▶ When figuring out split-complementary colors, you want to start out with a base color. From there, you combine it with two colors that sit directly

- **Primary colors** are colors you can't create by combining two or more other colors. The primary colors are red, blue, and yellow.
- **The secondary colors** are orange, purple, and green—in other words, colors that can be created by mixing any two of the three primary colors.
- **Tertiary colors** are created by mixing a primary color with a secondary color. The tertiary colors are magenta, vermillion, violet, teal, amber, and chartreuse.

JPEG

JPEG (Joint Photographic Experts Group) is a **lossy compression* method. Supports eight-bit grayscale images and 24-bit color images (eight bits each for red, green, and blue.) JPEG applies a lossy compression to images which can result in a significant reduction of the file size.

Keep in mind JPEG files suffer degradation when repeatedly edited and saved (it loses picture quality).

Recommended Use: When file size is more important than image quality. An example would be trying to save storage space on a website.

BMP

The **BMP file format** (Windows bitmap) handles graphic files within the Microsoft Windows OS. Typically, BMP files are uncompressed, and therefore large and lossless; their advantage is their simple structure and wide acceptance in Windows programs.

Recommended Use: Large files (uncompressed), was used mostly for printing.

GIF

GIF (Graphics Interchange Format) is for common usage limited to an 8-bit palette, or 256 colors (while 24-bit color depth is possible).

The GIF format is most effective for storing graphics with few colors, such as simple diagrams, shapes, logos and cartoon style images, as it

PNG

PNG (Portable Network Graphics) is a raster graphics **file** format that supports lossless data compression. **PNG** was created as an improved, non-patented replacement for **Graphics Interchange Format (GIF)**, and is the most used lossless image compression format on the Internet.

SVG

SVG (Scalable Vector Graphic) is the description of an image as an application of the Extensible Markup Language (XML). Any program such as a Web browser that recognizes XML can display the image using the information provided in the SVG format.

The SVG format enables the viewing of an image on a computer display of any size and resolution.

Recommended Use: Displaying images on screens (computers, lcd, smart phones, etc.)